Kevin Mekulu

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RESEARCH PROFILE

Industrial and systems engineer developing quantitative representations of human behavior for interpretable measurement and inference. My research combines stochastic and Markov modeling, nonlinear and recurrence analysis, and statistical learning with speech and gaze data. My independent program connects behavioral measurement with personalized monitoring and decisions under uncertainty in human-centered systems. Contributions span methodological innovation, biomarker discovery, inventions, and prospective human studies.

EDUCATION

Expected May 2027 Ph.D. Candidate, Industrial Engineering, The Pennsylvania State University
Dissertation: Multisensing AI for Early Cognitive Health: Speech, Eye-Tracking, and Digital Biomarkers
Advisor: Hui Yang, Ph.D.
May 2018 M.S., Applied Computational Mathematics and Statistics, University of Notre Dame
May 2017 B.S., Applied Mathematics, cum laude, Virginia Tech; NCAA Division I athlete

RESEARCH EXPERIENCE AND LEADERSHIP

2024-Present Research Team Lead, Complex Systems Monitoring, Modeling and Control Laboratory, Penn State

- Lead 12+ undergraduate researchers across engineering, computer science, neuroscience, and health sciences; spearheaded immersive VR environments for learning, manufacturing, and cognitive assessment.

2024-Present Founder & Chief Executive Officer, DementiAnalytics

- Develop and prospectively evaluate speech- and language-based measurement for cognitive health in senior-living environments, including a brief screening and monitoring system.
- Lead an IRB-approved prospective study with 49 participants with clinical ground-truth labels, approximately 294 recordings, and five enrolled senior-living communities; preparing longitudinal reassessment to evaluate within-person biomarker change.
- Coordinate biomarker research with speech-language pathology, neuropsychology, senior-living, and engineering collaborators, including Nicole Etter, Ph.D.; Timothy Brearly, Ph.D.; and Stephanie Ruth Young, Ph.D., Northwestern University Feinberg School of Medicine.

RESEARCH INTERESTS

Stochastic and Markov modeling; nonlinear dynamics and recurrence analysis; temporal behavioral measurement; interpretable statistical learning; speech and gaze; cognitive health. Future research: task-dependent and personalized inference, uncertainty quantification, longitudinal monitoring, sequential measurement decisions, and human-AI systems.

EXTERNAL FUNDING AND GRANT ACTIVITY

2024-2026 NSF National I-Corps Teams Award, $50,000, IIP-2425827
Artificial Intelligence for Dementia Assessment - **Kevin Mekulu, Co-Principal Investigator, Entrepreneurial Lead, and primary proposal writer**; Hui Yang, Principal Investigator and Technical Lead
2026 Submitted: NSF Translation to Practice - Translate Track (TTP-T), $1,200,000 - Co-PI and primary proposal writer

INNOVATION AND TRANSLATIONAL PROGRAMS

2024-2025 NIH I-Corps@NCATS Biomedical Translation Program
Penn State Center for Medical Innovation - **Kevin Mekulu, Principal Investigator and Entrepreneurial Lead** - Customer-discovery travel support

2023-2024 NSF I-Corps Mid-Atlantic Hub Award
$5,000 in customer-discovery support - **Kevin Mekulu, Principal Investigator and Entrepreneurial Lead**

PEER-REVIEWED JOURNAL PUBLICATIONS

1. **Mekulu, K.**, Aqlan, F., & Yang, H. (2026). A 60-second interpretable voice model for early dementia screening. *PLOS Digital Health*, 5(7), e0001552 https://doi.org/10.1371/journal.pdig.0001552
2. **Mekulu, K.**, Aqlan, F., & Yang, H. (2026). Agentic neurotechnology: A multi-agent AI framework for next-generation dementia care. *Alzheimer's & Dementia: Translational Research & Clinical Interventions*, 12, e70299 https://doi.org/10.1002/trc2.70299

3. **Mekulu, K.**, Aqlan, F., & Yang, H. (2026). A transparent four-feature speech model for depression screening applicable across clinical and community settings, including assisted-living environments. *Frontiers in Digital Health*, 7, 1675103 https://doi.org/10.3389/fdgth.2025.1675103
4. **Mekulu, K.**, Aqlan, F., & Yang, H. (2025). Character-level linguistic biomarkers for precision assessment of cognitive decline: A symbolic recurrence approach. *Frontiers in Aging Neuroscience*, 17, 1681124 https://doi.org/10.3389/fnagi.2025.1681124
5. **Mekulu, K.**, Aqlan, F., & Yang, H. (2025). CharMark: Character-level Markov modeling for interpretable linguistic biomarkers of cognitive decline. *Frontiers in Digital Health*, 7, 1659366 https://doi.org/10.3389/fdgth.2025.1659366
6. **Mekulu, K.**, Aqlan, F., & Yang, H. (2025). Agentic artificial intelligence in cognitive screening: A translational roadmap for dementia care. *Journal of Alzheimer's Disease Reports* https://doi.org/10.1177/25424823251407989
7. **Mekulu, K.**, Aqlan, F., & Yang, H. (2025). Reimagining dementia screening: A stakeholder-informed perspective on artificial intelligence, digital biomarkers, and real-world implementation. *Journal of Alzheimer's Disease Reports* https://doi.org/10.1177/25424823251395310
8. **Mekulu, K.**, Aqlan, F., & Yang, H. (2025). The mild cognitive impairment window for optimal Alzheimer's disease intervention. *Journal of Alzheimer's Disease Reports*, 9(1) https://doi.org/10.1177/25424823251370768

MANUSCRIPTS UNDER REVIEW OR REVISION

9. **Mekulu, K.**, Etter, N., Brearly, T., & Yang, H. Speech timing measures are not interchangeable across elicitation methods in senior living residents. Journal of Alzheimer’s Disease Reports (Major revision invited after peer review)

MANUSCRIPTS IN PREPARATION

10. **Mekulu, K.**, Begum, R., Aqlan, F., & Yang, H. From gaze transitions to digital biomarkers: Markov prediction of virtual-reality task performance. (Manuscript in preparation)
11. **Mekulu, K.**, Aqlan, F., & Yang, H. Nonlinear multifractality of gaze dynamics in immersive virtual reality. (Manuscript in preparation)

PEER-REVIEWED CONFERENCE PROCEEDINGS AND ABSTRACTS

12. **Mekulu, K.**, Aqlan, F., & Yang, H. (2025). Automated detection of early-stage dementia using large language models: A comparative study on narrative speech. *Alzheimer's & Dementia*, 21(S7), e108503 https://doi.org/10.1002/alz70861_108503
13. **Mekulu, K.**, Aqlan, F., & Yang, H. (2025). A novel AI-driven Markov chain approach to dementia detection. *Proceedings of the IISE Annual Conference*
14. **Mekulu, K.**, Aqlan, F., & Yang, H. (2024). AI and dementia diagnosis: Insights from linguistic patterns. *Proceedings of the IISE Annual Conference*
15. **Mekulu, K.**, Aqlan, F., & Yang, H. (2023). Intelligent detection of neurocognitive disorders based on physiological and speech signals in a virtual environment. *Proceedings of the IISE Annual Conference*
16. Iznurov, A., **Mekulu, K.**, Bello, K., et al. (2023). AI-driven virtual reality for mental health assessment. *Proceedings of the IISE Annual Conference*

PREPRINTS

17. **Mekulu, K.**, Aqlan, F., & Yang, H. (2025). Automated detection of early-stage dementia using large language models: A comparative study on narrative speech. *medRxiv* https://doi.org/10.1101/2025.06.06.25329081
18. **Mekulu, K.** (2025). AI-enhanced memory list assessment system: Multi-dimensional analysis and automated strategy detection for next-generation cognitive screening. *medRxiv* https://doi.org/10.1101/2025.10.02.25337192
19. **Mekulu, K.** (2025). AI-enhanced animal naming as a digital biomarker for early cognitive screening. *medRxiv* https://doi.org/10.1101/2025.11.10.25339942

PATENTS AND INTELLECTUAL PROPERTY

Five patent applications covering speech-, language-, and eye-tracking-based cognitive assessment technologies, including international application PCT/US2026/020992, filed March 26, 2026, with earliest priority dated April 7, 2025 (applicant: The Penn State Research Foundation).

1. **System and Method for Voice-Based Detection of Depression and Early Cognitive Impairment**
2. **System and Method for Analyzing Speech Patterns to Detect Neurodegenerative Cognitive Disorders**
3. **System and Method for Multi-Dimensional Analysis and Automated Strategy Detection for Next-Generation Cognitive Screening**
4. **Telecommunication Method and System for Cognitive Evaluation via Detected Visual Cognition**; U.S. nonprovisional application filed; claims priority to U.S. Provisional Application No. 63/898,219, filed October 13, 2025.
5. **System and Method for Scoring Framework for Verbal Fluency Tasks in Early Cognitive Screening**

TEACHING EXPERIENCE

Fall 2017 Co-Instructor, ACMS 30440: Probability and Statistics, University of Notre Dame
Fall 2021 Teaching Assistant, IE 453: Simulation Modeling for Decision Support, Penn State
Spring 2021 Teaching Assistant, IE 330: Engineering Analytics, Penn State

TEACHING INTERESTS

Probability and statistics; stochastic processes and simulation; engineering data analytics; time-series analysis; interpretable machine learning; quantitative behavioral measurement; healthcare systems engineering. Proposed graduate elective: Behavioral Measurement and Interpretable AI for Human Systems.

HONORS AND FELLOWSHIPS

2026 MedTech Ahead Startup Program, Fall 2026 - DementiAnalytics selected from 100+ applicants
2026 Leighton Riess Graduate Fellowship in Engineering, Penn State College of Engineering
2025 Forbes 30 Under 30 - Healthcare
2025 Industrial and Manufacturing Engineering Doctoral Student Fellowship, $10,000, Penn State
2025 Samsung NEXT Mobile Advance Finalist, selected from 500+ global applicants
2025 Penn State AI Society Award, Social Impact Category
2024-2025 Diefenderfer Fellowship for Researcher Entrepreneurs, Penn State College of Engineering (two-time recipient)
2024 Ben Franklin Technology Partners Startup Accelerator Finalist, $1,000 award
2024 Microsoft for Startups Azure Credits, $5,000
2023 Penn State Graduate Research Exhibition, 2nd Place, Engineering Category

MENTORSHIP AND LEADERSHIP

- Mentor 12+ undergraduate researchers working across engineering, computer science, neuroscience, and health sciences.
- Led 180+ stakeholder interviews through NSF I-Corps with clinicians, caregivers, senior-living operators, and technology stakeholders.

PROFESSIONAL SERVICE

- Reviewer, IEEE Journal of Biomedical and Health Informatics
- Reviewer, Frontiers in Dementia - Dementia Care
- Reviewer, Data & Policy

SELECTED PRESENTATIONS AND INVITED TALKS

2025 Alzheimer's Association International Conference (AAIC) - "Automated Detection of Early-Stage Dementia Using Large Language Models"
2025 IEEE BHI - "Character-Level Linguistic Biomarkers for Precision Assessment of Cognitive Decline"
2025 Penn State Center for Socially Responsible AI - "The Cure is Timing: How AI Voice Biomarkers Open the Treatment Window in Alzheimer's"
2025 IISE Annual Conference, Atlanta, GA - "A Novel AI-Driven Markov Chain Approach to Dementia Detection"
2024 IISE Annual Conference, Montreal, Canada - "AI and Dementia Diagnosis: Insights from Linguistic Patterns"

SELECTED MEDIA RECOGNITION AND PUBLIC IMPACT

2026 Penn State Research Q&A on AI, Cognition, and Dementia [Link]
2025 Forbes 30 Under 30 - Healthcare [Link]
2024 Penn State Profile on NSF I-Corps and Dementia Research Translation [Link]
2024 McKnight's Senior Living Coverage of Multimodal Cognitive-Screening Research [Link]